In [20]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [21]:
```
df = pd.read_csv(r"C:\Users\tusha\OneDrive\Desktop\archive\Titanic_train.csv")
df.head()
```

Out[21]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	F
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0

In [22]:
```
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

Out[22]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.00
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.20
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.69
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.00
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.91
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.45
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.00
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.32

In [23]:
```
df.isnull().sum()
```

Out[23]:
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

In [24]:
```
# Missing values fix
df["Age"] = df["Age"].fillna(df["Age"].mean())
df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])

# Drop Cabin safely
df = df.drop("Cabin", axis=1, errors="ignore")
```

```
# Drop unnecessary columns
df = df.drop(["Name", "Ticket", "PassengerId"], axis=1, errors="ignore")

# Convert categorical to numeric
df = pd.get_dummies(df, columns=["Sex", "Embarked"], drop_first=True)

# Check
print(df.head())
print(df.isnull().sum())
```

```
   Survived  Pclass   Age  SibSp  Parch     Fare  Sex_male  Embarked_Q  \
0         0       3  22.0      1      0   7.2500      True       False
1         1       1  38.0      1      0  71.2833     False       False
2         1       3  26.0      0      0   7.9250     False       False
3         1       1  35.0      1      0  53.1000     False       False
4         0       3  35.0      0      0   8.0500      True       False

   Embarked_S
0        True
1       False
2        True
3        True
4        True
Survived      0
Pclass        0
Age           0
SibSp         0
Parch         0
Fare          0
Sex_male      0
Embarked_Q    0
Embarked_S    0
dtype: int64
```

In [25]:
```
from sklearn.model_selection import train_test_split

X = df.drop("Survived", axis=1)
y = df["Survived"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

In [26]:
```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

Out[26]:
```
▾ LogisticRegression
▶ Parameters
```

In [27]:
```
y_pred = model.predict(X_test)
```

In [28]:
```
from sklearn.metrics import accuracy_score

print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
Accuracy: 0.8100558659217877
```

In [29]:
```
from sklearn.metrics import precision_score, recall_score

print("Precision:", precision_score(y_test, y_pred))
print("Recall:", recall_score(y_test, y_pred))
```

```
Precision: 0.7857142857142857
Recall: 0.7432432432432432
```

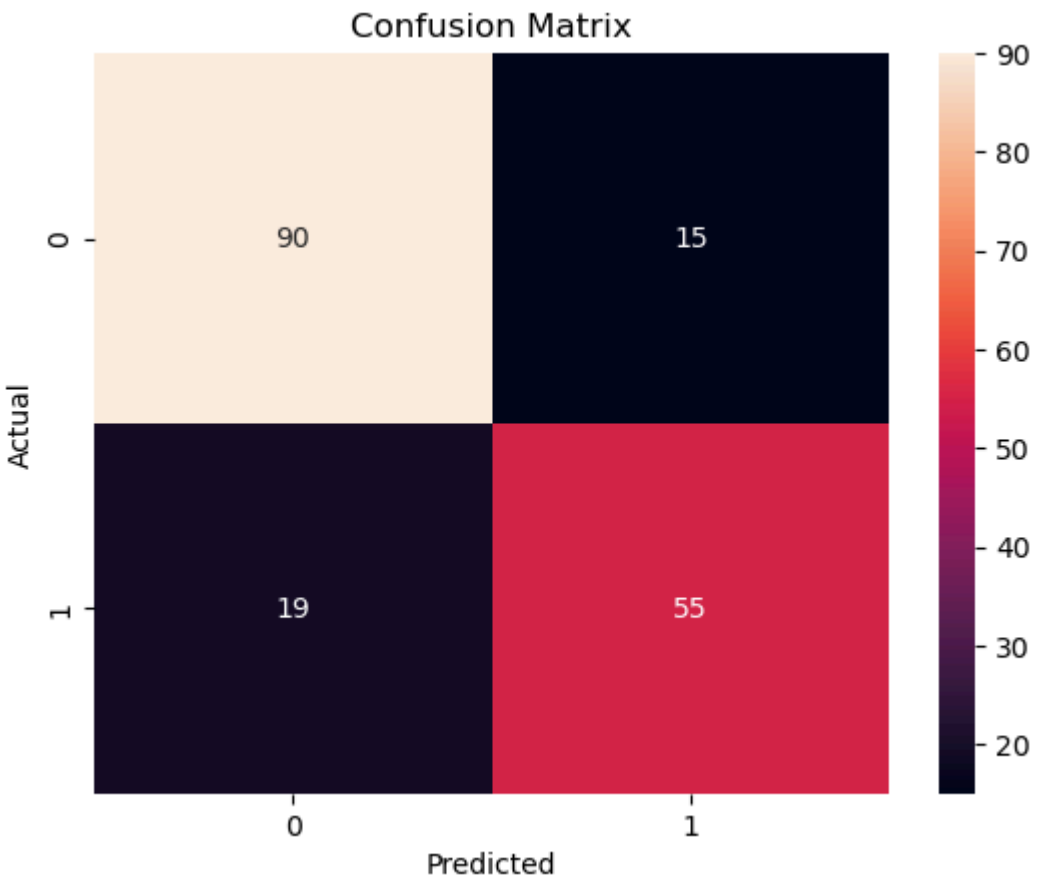
In [30]:
```
from sklearn.metrics import confusion_matrix

cm = confusion_matrix(y_test, y_pred)
print(cm)
```

```
[[90 15]
 [19 55]]
```

In [31]:
```
import seaborn as sns
import matplotlib.pyplot as plt

sns.heatmap(cm, annot=True, fmt='d')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```



In [32]:
```
from sklearn.metrics import classification_report

print(classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       0.83      0.86      0.84       105
           1       0.79      0.74      0.76        74

    accuracy                           0.81       179
   macro avg       0.81      0.80      0.80       179
weighted avg       0.81      0.81      0.81       179
```

In [ ]:

In [ ]: